



The 66th ASH Annual Meeting Abstracts

POSTER ABSTRACTS

721.ALLOGENEIC TRANSPLANTATION: CONDITIONING REGIMENS, ENGRAFTMENT, AND ACUTE TOXICITIES

Lower Respiratory Tract Pathogen Colonization Detected By Metagenomic Next-Generation Sequencing of Bronchoalveolar Lavage Fluid before Allogeneic Hematopoietic Stem Cell Transplantation and Its Predicting Value for Post-Transplant InfectionHang Zhang, MD¹, Hui Liu, MD², Ting Liu³, Ting Niu, MDPH⁴, Zhigang Liu, MD PhD⁵, Jie Ji, MD PhD⁶¹Department of Hematology and Institute of Hematology, West China Hospital, Sichuan University, Chengdu, China²Department of Hematology, People's Hospital of Guang'an City, Guang'an, China³Department of Hematology, West China Hospital of Sichuan University, Chengdu, China⁴West China Hospital, Sichuan University, Chengdu, China⁵Department of Hematology, West China Hospital, Sichuan University, Chengdu, China⁶Department of Hematology and Institute of Hematology, Department of Hematology, West China Hospital of Sichuan University, Chengdu, China**Introduction**

Pulmonary infection remains a leading cause of early mortality following allogeneic hematopoietic stem cell transplantation (allo-HSCT). Previous studies have demonstrated that metagenomic next-generation sequencing (mNGS) of bronchoalveolar lavage fluid (BALF) can rapidly and accurately detect pathogens in patients with pulmonary infection. This study investigates the predictive value of mNGS of BALF before allo-HSCT for post-transplant infections.

Methods

This prospective, single-arm clinical trial involved patients scheduled for allo-HSCT. Patients underwent bronchoalveolar lavage (BAL) with fiberoptic bronchoscopy under general anesthesia before transplantation. BALF samples were subjected to perform mNGS. Patients subsequently received allo-HSCT according to standard clinical protocols. mNGS results were considered positive if pathogenic or opportunistic pathogens were detected. Post-transplant pulmonary infections were diagnosed by two experienced clinicians based on clinical symptoms, imaging examinations, microbiological tests, and inflammatory markers.

Results

A total of 33 patients (19 males, 14 females) with a median age of 41 years (range 15–61) were enrolled. 24 patients were diagnosed with acute leukemia, 6 with myelodysplastic syndrome, 2 with lymphoma, and 1 with chronic myeloid leukemia. No patient exhibited respiratory infection symptoms, nor BAL- or general-anesthesia-related adverse reactions when performing BAL. mNGS reported pathogenic or opportunistic organisms in 22 patients (66.7%).

The overall rate of bacterial colonization was 36.4% (12/33), with *Pseudomonas aeruginosa* (9.1%) and *Acinetobacter baumannii* (9.1%) being the most common, other including *Moraxella catarrhalis*, *Haemophilus influenzae*, *Tropheryma whipplei*, *Klebsiella pneumoniae*, *Mycobacterium tuberculosis* complex, *Mycobacterium abscessus*, *Morganella morganii*, *Serratia marcescens*, and *Staphylococcus aureus*. Fungal colonization was observed in 36.4% of patients (12/33), predominantly *Pneumocystis jirovecii* (27.3%), and sporadically *Aspergillus* spp. (9.1%), *Mucor* spp. (3.0%), *Cryptococcus neoformans*, *Candida albicans*, and *Candida tropicalis*. Viral colonization was detected in 9 patients (27.3%), including human herpesvirus 6B (12.1%), Epstein-Barr virus (9.1%), human respiratory syncytial virus type 3 (6.1%), SARS-CoV-2 (3.0%), cytomegalovirus (3.0%), influenza B virus (3.0%), and adenovirus type 3 (3.0%).

Patients underwent allo-HSCT subsequently, and 24 reached 100 days post-transplant (16 mNGS positive and 8 mNGS negative). Twenty patients received stem-cell grafts from sibling haploidentical donors, while 4 from matched sibling donors. Acute graft-versus-host-disease prophylaxis included cyclophosphamide and cyclosporine, with mycophenolate mofetil added for haploidentical allo-HSCT. The median time to neutrophil and platelet engraftment was 14 days (range 10–24) and 17 days (range 12–54), respectively. The median follow-up duration was 6.0 months (range 1.3–9.0 months). By day +30 post-ASCT, all patients achieved complete remission and minimal residual disease clearance with full donor chimerism. Of the 16 mNGS-positive patients, 9 (56.3%) developed infections within 100 days post-transplant, and 2 of them died of it. One patient presented with mixed infections (pulmonary fungal + viral infection and intestinal fungal infection). Additionally, 2 patients de-

veloped bacterial infections (one pulmonary and one urinary tract infection), 4 patients had pulmonary fungal infections, and 2 patients experienced pulmonary viral infections. None of the 8 mNGS-negative patients developed pulmonary infections within 100 days, though 1 patient developed pulmonary fungal infection on day +108 and died on day +162. Two patients exhibited bloodstream *E. coli* infection within 100 days and died of it. The rate of infection had no significant difference ($P=0.211$). The rate of 6-month infection-related mortality was 41.7% in mNGS-positive population, and 12.5% in mNGS-negative population ($P=0.315$).

Conclusion

Pre-transplant BAL is safe for allo-HSCT patients, and lower respiratory tract fungal colonization is common in this population. Further research is needed to determine whether decolonization and infection prevention strategies are necessary for patients with pathogen colonization.

Disclosures No relevant conflicts of interest to declare.

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